

Homework 7

BSTA 551

Directions

Same as previous instructions.

For simulation problems, set your seed to `set.seed(551)` for reproducibility.

For the qmd of this Homework page, see this [github link](#)

You must show all of your work to receive credit.

This homework covers material from Lessons 18, 19, and 20: Two-Sample Inference (z, t, proportions, F-test), Analysis of Paired Data, and Bootstrap/Permutation Methods for Two Samples.

Questions

Problem 1: Two-Sample t-Tests for Means (Lesson 18 — Devore Section 9.5)

A pulmonologist is comparing plasma fibrinogen levels (mg/dL) between patients diagnosed with COPD and age-matched healthy controls. High fibrinogen is associated with systemic inflammation. The following summary statistics were recorded:

Group	n	$\bar{x}$	s
COPD	22	412.8	68.4
Healthy controls	25	356.2	41.7

Assume the data within each group are approximately normally distributed.

- (a) Compute the Welch–Satterthwaite approximation to the degrees of freedom $\hat{\nu}$ by hand (you may verify using R). Show all intermediate calculations.
- (b) Test $H_0 : \mu_1 - \mu_2 = 0$ vs. $H_a : \mu_1 > \mu_2$ at $\alpha = 0.05$ using the Welch t -statistic. State the test statistic, approximate degrees of freedom, p -value (use R to compute), and conclusion.
- (c) Compute a 95% two-sided confidence interval for $\mu_{\text{COPD}} - \mu_{\text{Control}}$. Interpret in the context of this study.
- (d) Now test whether the population variances are equal using the F -test. State the hypotheses, the test statistic $F_0 = s_1^2/s_2^2$, degrees of freedom, and p -value. Based on this result, is the Welch t -test or pooled t -test more appropriate? Explain briefly.
- (e) For reference, carry out the pooled t -test as well. Report the pooled estimate S_p^2 , the pooled test statistic, and the 95% CI. How do the Welch and pooled results compare? Under what condition on n_1, n_2, s_1, s_2 might they differ substantially?

Hint: You do not have raw data, so use `pt()` and `qchisq()` / `qf()` as needed. You can construct a function to compute the Welch-Satterthwaite df from the summary statistics.

Problem 2: Two-Sample Inference for Proportions (Lesson 18 — DBC 9.6)

A multi-site clinical trial randomizes 210 patients with community-acquired pneumonia to one of two antibiotic regimens:

- **Regimen A:** 105 patients, 89 classified as treatment successes at Day 14
- **Regimen B:** 105 patients, 76 classified as treatment successes at Day 14

Let p_A and p_B denote the true success probabilities.

- (a) Verify that the large-sample normal approximation is appropriate for inference on $p_A - p_B$ (check all four conditions).
- (b) Compute a 95% confidence interval for $p_A - p_B$ using the **unpooled** standard error. Interpret the interval in the context of the study. Do these data provide evidence that the two regimens differ in success rate?
- (c) Test $H_0 : p_A = p_B$ vs. $H_a : p_A \neq p_B$ at $\alpha = 0.05$ using the **pooled** standard error. Show your calculation of $\hat{p}_{\text{pool}}$, the test statistic Z , and the p -value. Verify using `prop.test()` with `correct = FALSE`.
- (d) Explain why the test in part (c) uses the **pooled** estimate $\hat{p}_{\text{pool}}$ while the CI in part (b) uses the **unpooled** SEs. What assumption does H_0 impose that justifies pooling?

(e) Suppose the trial statistician also wants to estimate the **relative risk** $RR = p_A/p_B$. Compute the point estimate. Using the delta method (or a log-transform approach), give a 95% CI for RR.

Hint for (e): Apply the delta method to $\log(\hat{p}_A/\hat{p}_B)$. If $\hat{p}$ is a sample proportion, $\text{Var}(\log \hat{p}) \approx (1-p)/(np)$ by the delta method.

Problem 3: Paired Data Analysis (Lesson 19 — DBC 10.3)

A clinical investigator studies the effect of a 16-week low-glycemic-index dietary intervention on insulin resistance. Insulin resistance is measured via the **HOMA-IR** index (higher values = more insulin resistance) in 15 participants, both before and after the intervention. The individual HOMA-IR values are:

```
before <- c(3.8, 5.2, 4.1, 6.7, 3.3, 4.9, 5.8, 7.1, 3.6, 4.4,
            6.0, 5.5, 4.7, 3.9, 5.1)
after  <- c(2.9, 4.4, 3.2, 5.8, 2.8, 4.1, 4.9, 6.0, 2.7, 3.5,
            5.0, 4.6, 3.8, 3.1, 4.3)
```

(a) Explain why a **paired** analysis is appropriate here rather than an independent two-sample test. What would be the consequences of incorrectly applying the Welch two-sample t -test?

(b) Compute the within-subject differences $D_i = \text{Before}_i - \text{After}_i$. Report $\bar{D}$, S_D , and the sample correlation r between the before and after measurements.

(c) Using the paired t -test, test $H_0 : \mu_D = 0$ vs. $H_a : \mu_D > 0$ (intervention reduces HOMA-IR). Report the test statistic, p -value, and conclusion at $\alpha = 0.05$.

(d) Compute a 95% two-sided confidence interval for μ_D . Interpret in context.

(e) Now apply the (incorrect) Welch two-sample t -test to the same data by ignoring the pairing. Report the p -value and 95% CI. Compare the SE of the estimate, the width of the CI, and the p -value to the paired analysis. Explain quantitatively why they differ using the variance formula from the slides:

$$\sigma_D^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

(f) Construct a spaghetti plot (individual patient profiles, before vs. after) and a histogram of the differences D_i . Add a vertical line at $D = 0$ and annotate $\bar{D}$. Use `ggplot2`.

Problem 4: Bootstrap and Permutation Inference (Lesson 20 — C&H Ch. 3, 5, 8)

A hematologist is comparing **platelet counts** ($\times 10^3/\mu\text{L}$) between patients admitted with **septic shock** ($n_1 = 35$) and **non-septic sepsis** ($n_2 = 40$) to a medical ICU. Platelet counts in critically ill patients are often right-skewed with heavy tails. Use the following code to generate the data:

```
set.seed(551)
platelets_shock <- round(rgamma(35, shape = 2.0, rate = 0.016), 1)
platelets_sepsis <- round(rgamma(40, shape = 3.5, rate = 0.022), 1)
```

(a) Produce exploratory summaries: compute the mean, median, and SD for each group and create side-by-side density plots (or violin plots) overlaid with rug marks. Comment on the distributional shape of each group.

(b) Compute a 95% bootstrap **percentile** CI and a 95% bootstrap **basic (reflection)** CI for the difference in means $\mu_{\text{shock}} - \mu_{\text{sepsis}}$ using $B = 5000$ bootstrap resamples. Report both CIs and the bootstrap SE.

(c) Compute a 95% bootstrap **percentile** CI for the difference in **medians** $\tilde{\mu}_{\text{shock}} - \tilde{\mu}_{\text{sepsis}}$ using $B = 5000$ resamples. Report the point estimate and CI.

(d) Compute a two-sided Welch t -test and 95% CI for the difference in means. Compare to the bootstrap CI from (b). In what way do the bootstrap and t CIs agree or disagree? What does this tell you about the appropriateness of the t approximation here?

(e) Perform a **two-sample permutation test** of $H_0 : F_1 = F_2$ vs. $H_a : F_1 \neq F_2$ using the difference in sample means as the test statistic with $B = 5000$ permutations. Report the observed test statistic, the permutation p -value, and compare it to the Welch t -test p -value. Under what null hypothesis is the permutation test *exactly* valid?

(f) Visualize the permutation null distribution. Plot a histogram of the $B = 5000$ permuted statistics, add a vertical line at the observed statistic, and shade the area corresponding to the p -value. Use `bquote()` or `expression()` for axis labels involving statistical notation.

add to final exam

Problem 5: Conceptual Questions

(a) A researcher has $n = 12$ matched pairs and the differences $D_i = X_{1i} - X_{2i}$ appear moderately right-skewed. She applies the paired t -test. Discuss: (i) whether the test is reliable in this setting; (ii) what alternatives she might consider; and (iii) how bootstrap inference on the paired differences could be applied.

(b) Explain why the two-sample permutation test tests $H_0 : F_1 = F_2$ (the two *distributions* are equal) rather than just $H_0 : \mu_1 = \mu_2$ (the two means are equal). Under what additional assumptions would the permutation test provide a valid test of the location hypothesis $H_0 : \mu_1 = \mu_2$?

(c) In the F-test for $H_0 : \sigma_1^2 = \sigma_2^2$, what happens to the Type I error rate when the underlying populations are non-normal (e.g., exponential or t_3)? How does Levene's test address this limitation?